

III - Mean Value Theorems of differential Calculus.

Title refers to a group of theorems that are of paramount importance.

All these follow from Rolle's Theorem which in turn is a simple consequence of the fundamental result:

Thm B on p32 and its Supplement Thm D on p34. This is precisely the reason for discussing in detail the proofs on p32 ff.

Thm (Rolle's Thm): Suppose $f: [a, b] \rightarrow \mathbb{R}$ is continuous, and differentiable on (a, b) . Further, assume $f(a) = f(b)$. Then there exists a $c \in (a, b)$ (open interval!) such that $f'(c) = 0$.

proof: $M = \text{l.u.b } \{ f(x) / x \in [a, b] \}$
 $m = \text{g.l.b } \{ f(x) / x \in [a, b] \}$

$M = f(p) ; m = f(q)$ for $p, q \in [a, b]$

if p, q are both endpoints of $[a, b]$

($p = a, q = b$ or $q = a, p = b$) then by virtue of $f(a) = f(b)$ we conclude

at $M = m$ and f is const on $[a, b]$ and $\therefore$ Every point in (a, b) , f' vanishes ✓

Assume p or q is in the open interval (a, b) then the derivative at that point is 0

Assume $p \in (a, b)$: $f(p) = M = \sup \{ f(x) / a \leq x \leq b \}$
 p is a point of local Max and so $f'(p) = 0$
 (See p 41) . Rolle's thm is proved

Cor 1: Suppose $f, g, h : [a, b] \rightarrow \mathbb{R}$ cont.
 and differentiable on (a, b) then there
 exists $c \in (a, b)$ such that

$$\begin{vmatrix} f'(c) & g'(c) & h'(c) \\ f(a) & g(a) & h(a) \\ f(b) & g(b) & h(b) \end{vmatrix} = 0. \quad (*)$$

proof: Let $\phi(x) = \begin{vmatrix} f(x) & g(x) & h(x) \\ f(a) & g(a) & h(a) \\ f(b) & g(b) & h(b) \end{vmatrix}$

$$\phi(a) = \phi(b) = 0$$

ϕ is cont. on $[a, b]$ and diff. on (a, b)

By Rolle's thm there is a $c \in (a, b)$ s.t.
 $\phi'(c) = 0$.

Exercise: Check that $\phi'(c) = 0$ is precisely (*)

Cor 2: Cauchy's MVT (CMVT) Suppose
 $f, g : [a, b] \rightarrow \mathbb{R}$ cont. diff on (a, b) . Then
 there exists $c \in (a, b)$ such that

$$f'(c)(g(b) - g(a)) = g'(c)(f(b) - f(a))$$

pf: Take $h(x) = 1$ in Cor 1 (Exercise)

Cor 3 Lagrange's MVT (LMVT): If
 $f : [a, b] \rightarrow \mathbb{R}$ continuous and diff on (a, b)
 then

There exists $c \in (a, b)$ s.t.

$$f(b) - f(a) = f'(c)(b-a)$$

pf: Exercise. Take $g(x) = x$ in CMVT

Cor 4: If $f: (a, b) \rightarrow \mathbb{R}$ is differentiable and $f'(x) > 0$ throughout (a, b) then f is strictly increasing.

That is $x, y \in (a, b)$ with $x < y$
 $\Rightarrow f(x) < f(y)$

pf: Apply LMVT to f on $[x, y]$

Formulate a parallel result when $f'(x) < 0$ throughout (a, b)

Cor 5: (L'Hospital's Rule)

Suppose $f, g: (a, b) \rightarrow \mathbb{R}$ differentiable

and $p \in (a, b)$ such that $g(x) \neq 0$ if $x \neq p$

(i) $f(p) = g(p) = 0$: $g'(x) \neq 0$ on (a, b)

(ii) $\lim_{x \rightarrow p} \frac{f'(x)}{g'(x)}$ exists $= l$

Then

$\lim_{x \rightarrow p} \frac{f(x)}{g(x)}$ exists and $= l$.

Note: We need (ii) ~~division~~ to hold only on some interval $(p-c, p+c)$

That $g'(x)$ could vanish at points in (a, b) as long as we are guaranteed $g(x)$ and $g'(x) \neq 0$ in an interval $(p-c, p+c)$

For computing limits and derivatives of functions at a point p all that matters is behaviour in some interval $(p-c, p+c)$

Proof: We have to prove that
if (x_n) is a seq of points such
that $x_n \rightarrow p$ and $x_n \neq p$
the corresp. sequence $\frac{f(x_n)}{g(x_n)} \rightarrow l$.

$$\text{Well, } \frac{f(x_n)}{g(x_n)} = \frac{f(x_n) - f(p)}{g(x_n) - g(p)} = \frac{f'(c_n)}{g'(c_n)}$$

by CMVT:

c_n lies between p and x_n

So, $|c_n - p| \leq |p - x_n|$ but $|p - x_n| \rightarrow 0$

$\therefore c_n \rightarrow p$.

Since $\lim_{x \rightarrow p} \frac{f'(x)}{g'(x)}$ is given to exist and $= l$.

$$\text{We conclude } \frac{f'(c_n)}{g'(c_n)} \rightarrow l$$

$\therefore \frac{f(x_n)}{g(x_n)} \rightarrow l$. Then is proved!

Exercises: If $P(x)$ is a polynomial

a root c is called a double simple

root if $P(x) = (x-c)Q(x); Q(c) \neq 0$

c is called a double root if

$$P(x) = (x-c)^2 Q(x); Q(c) \neq 0.$$

c is called a triple root if

$$P(x) = (x-c)^3 Q(x); Q(c) \neq 0 \text{ etc;}$$

Prove that if c is a k -fold root of $p(x)$ namely (Polynomial)

$$p(x) = (x-c)^k Q(x): Q(c) \neq 0$$

then

$$\left. \begin{aligned} p(c) = 0, p'(c) = 0, \dots, p^{(k-1)}(c) = 0 \\ \text{and } p^{(k)}(c) \neq 0 \end{aligned} \right\} (*)$$

Conversely if $(*)$ holds then c is a k -fold root of $p(x)$

Exercise: Suppose f is a polynomial such that a, b are both double roots of f ($a < b$) then $f''(x)$ vanishes at least at two distinct points in (a, b) .

Exercise: the prove that $\left(\frac{d}{dx}\right)^n (x^2-1)^n$ (**) is a polynomial

has n distinct roots in $(-1, 1)$

Hint: $f(a) = f(b) = 0$

Apply Rolle's thm to get $f'(c) = 0; a < c < b$

Apply Rolle's thm again on (a, c) and (c, b) (to what function?)

The polynomial $(**)$ has degree n and so cannot have more than n roots. With the factor $\frac{1}{2^n n!}$ it is denoted by $P_n(x)$:

$$P_n(x) = \frac{1}{2^n n!} D^n (x^2-1)^n \quad (\text{Legendre Poly}).$$

Sufficient Cond. for a point to be a local
Max or a local Min: I is an open interval
 $f: I \rightarrow \mathbb{R}$ differentiable

$$f'(p) = 0 : p \in I.$$

$c > 0$ is such that $(p-c, p+c) \subset I$

We shall put further cond. to ensure p is a point of
local Max/Min

Thm:

Suppose $f'(x) > 0$ on $(p-c, p)$ } $\Rightarrow p$ is a pt
 $f'(x) < 0$ on $(p, p+c)$ } of local Max (i)

Suppose $f'(x) < 0$ on $(p-c, p)$ } $\Rightarrow p$ is a pt of
 $f'(x) > 0$ on $(p, p+c)$ } local Min (ii)

We shall prove (i)

proof of (ii) Exercise for the student.

By Cor 4 on p. 44

f is strictly inc on $(p-c, p)$

In particular, let $x \in (p-c, p)$; $x < p$

$\therefore x < p - \frac{1}{n}$ for all n sufficiently large

$$\therefore f(x) < f(p - \frac{1}{n}) \dots$$

But $p - \frac{1}{n}$ monotone inc. to p

Continuity of f now implies the seq

$$f(p - \frac{1}{n}) \text{ monotone inc } \& \rightarrow f(p)$$

$$\therefore f(x) < f(p)$$

Thrs $f(x) < f(p)$ For all $x \in (p-c, p)$

Next f is strictly dec. on $(p, p+c)$

Let $x \in (p, p+c)$

$p + \frac{1}{n} < x$ for all n sufficiently large

$p + \frac{1}{n}$ monotone $\downarrow$ and conv. to p .

$\therefore f(p + \frac{1}{n})$ is monotone increasing
and Converges to $f(p)$
 $f(x) < f(p + \frac{1}{n}) \rightarrow f(p)$ also
 $\therefore f(x) < f(p)$ for $x \in (p, p+c)$
 $\therefore f(x) < f(p) \quad \forall x \in (p-c, p+c), x \neq p$
 p is a point of local Max

Cor : If $f: I \rightarrow \mathbb{R}$ is twice differentiable
and $f'(p) = 0$.

$f''(p) > 0 \Rightarrow p$ is a point of local Min
 $f''(p) < 0 \Rightarrow p$ is a point of local Max

Well, $f''(p) < 0 \Rightarrow f'$ is strictly dec.
at p so there is a $c > 0$ s.t.
 $(p-c, p+c) \subset I$ and

$$f'(x) < f'(p) = 0 \quad \text{on } (p, p+c)$$

$$f'(x) > f'(p) = 0 \quad \text{on } (p-c, p)$$

By the Main-thm (p 47) we get that p is
a point of local Max.

The Other Case $f''(p) > 0$ left for the
Student to work out...

(Reading Ex for Student $\downarrow$)

Remark: In many problems one computes
the roots of $f'(x) = 0$

They are usually finite in number say
 p_1, p_2, p_3 for example.

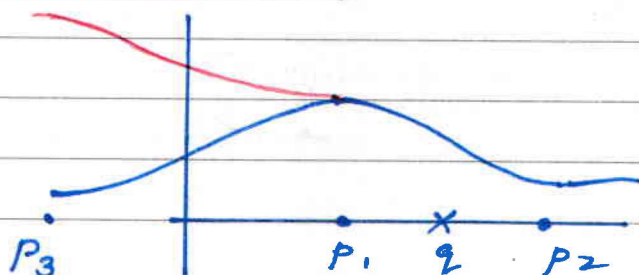
$$f'(p_1) = 0; f'(p_2) = 0; f'(p_3) = 0$$

There is No need to compute $f''(p)$

-49-

BECAUSE, f' maintains the same sign between p_1 and p_2

One can simply evaluate $f'(q)$ at some convenient point $q \in (p_1, p_2)$ and figure out the sign of f' - here if $f' < 0$ on (p_1, p_2) - then f must decrease from p_1, p_2



Now check the sign of $f'(q)$ for some convenient q in (p_3, p_1)

If this sign is +ve then we know f increases on (p_3, p_1) (blue curve)

However if this sign is -ve - then f decreases from p_3 to p_1 (~~Red curve~~ Arc)

p_1 is a maxima for the ^{completely} blue curve

p_1 is neither a Max nor a Min for the Red-blue curve

This is useful because computing the second derivative can be quite a tedious task.

Try $f(x) = e^{-x^2} \left(\frac{1-x^2}{1+x^2} \right)$. Is origin a Max/Min?

- 50 -

Polynomial
Compute this Expression

$$f'(x) = 2x Q(x) (\dots)$$

$(1+x^2)^2$ clubbed with $Q(x)$

$Q(x)$ is never vanishing and always positive

$$f'(x) = 2x Q(x) \text{ (positive)}$$

$$\text{Sign } f'(x) = \text{Sign } x(\dots)$$

$2 Q(x)$ has been ignored !

Note that Computing $f''(x)$ would be quite troublesome

(End of Reading Ex)

Points of Inflection (In older books the

Spelling is inflexion: G. A. Gibson, elementary treatise on the Calculus)

For a twice differentiable function

$f: I \rightarrow \mathbb{R}$, a point $p \in I$ is said to be a point of inflection if $f''(p) = 0$, $f'''(p) \neq 0$

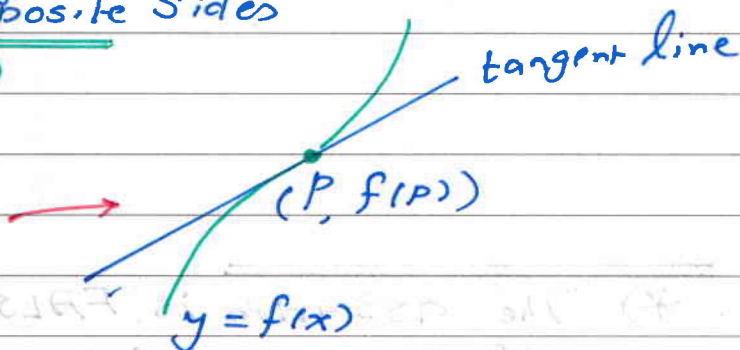
(Thomas-Finney, 9th Ed, page 211)

Geometrically, at the point $(p, f(p))$ on the graph, the graph crosses its tangent and bends away from it in opposite directions on opposite sides

(p 35 of G. A. Gibson)

$$f''(p) = 0 \text{ and } f'''(p) \neq 0.$$

$$\text{E.g.: } f(x) = x^3$$



Thomas Finney (9th Edition) p 219)

Problem 76: $f'(x) = (x-1)^2(x-2)(x-4)$

locate the points of local Max/Min

Does the function have points of inflection?

Problem 79: For what value of b will
 $x^3 + bx^2 + cx + d$ have an inflection
point at $x=1$?

Exercise: Let I be an open interval
and $f: I \rightarrow \mathbb{R}$

have a unique p at which $f'(p) = 0$.

Suppose that $f''(p) > 0$

We know p is a point of local Min

Can we assert that

$$f(p) = \inf \{ f(x) / x \in I \}$$

Would Rolle's thm help?

The Case with functions of several variables
is more interesting* and we shall return to
this in a later Chapter

*) The assertion is FALSE for functions
of several variables.

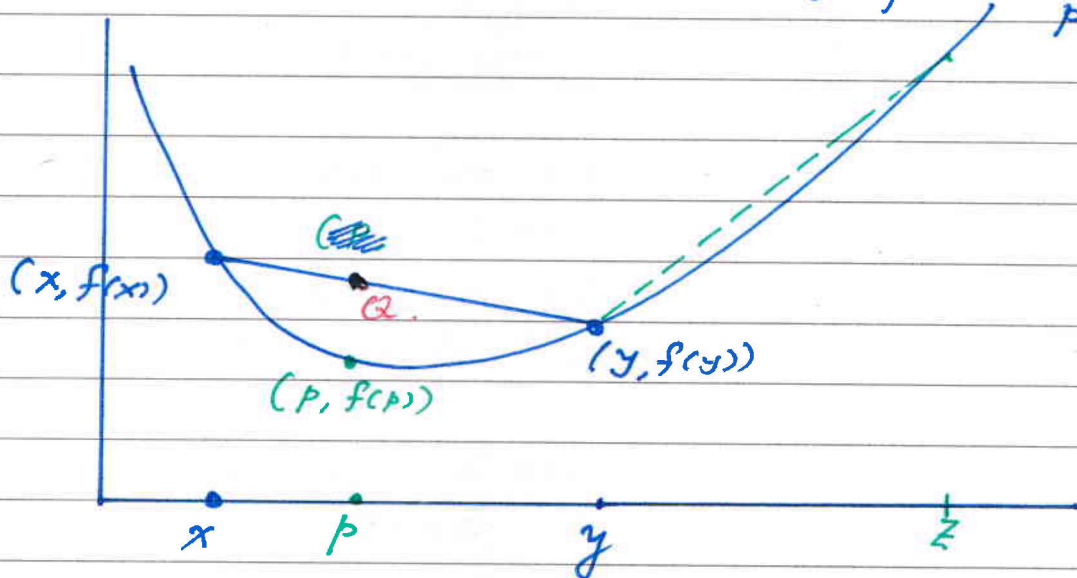
Convex functions:

I is an open interval.

$f: I \rightarrow \mathbb{R}$ is said to be convex if
 given $x, y \in I$
 $f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)$ (*)

In other words, the Chord joining
 $(x, f(x))$ and $(y, f(y))$ lies above
the graph of f

The nicest example is the parabola
 (slightly skewed in this pic)



$$p = tx + (1-t)y$$

Using say section formula:

$$Q = (p, tf(x) + (1-t)f(y))$$

Condition (*) simply says the point

Q lies above the point $(p, f(p))$

Basic Slope Lemma:

Let $f: I \rightarrow \mathbb{R}$ be convex

Assume $x < y < z$; $x, y, z \in I$

Slope Chord joining $(x, f(x))$ and $(y, f(y))$
 $\leq$ Slope Chord joining $(y, f(y))$ and $(z, f(z))$

(See the picture on p 52)

Exercise (Tutorial) Prove the basic
Slope Lemma

You have nothing but (*) and section
formula to play with!

How do these two slopes Compare with
Slope of Chord joining $(x, f(x))$, $(z, f(z))$
Can you make a guess? Prove that your
Guess is Correct.

Use the basic Slope Lemma to prove that
for $f: I \rightarrow \mathbb{R}$ differentiable,

f is Convex iff f' is increasing.

(In one direction LMVT will help)

Deduce that if $f: I \rightarrow \mathbb{R}$ is twice
differentiable

f is Convex iff $f'' \geq 0$.

Example: e^x is Convex

$-\log x$ is Convex on $(0, \infty)$

e^{-x} is Convex

- $\sin x$ is Convex on $(0, \pi/2)$

$\sin x$ is Concave on $(0, \pi/2)$ The chord joining two points lies below the graph

Ex: prove that $\sin x \geq \frac{2}{\pi} x$
for $0 < x < \pi/2$

(Draw the picture. Use Concavity)

What is $\lim_{R \rightarrow \infty} \int_0^{\pi/2} e^{-R \sin \theta} d\theta$?

Exercises:

Which of the following functions are Convex ?

- (i) $\log \sec \theta$ on $(-\pi/2, \pi/2)$
- (ii) $\log (e^x + e^{-x})$ on $\mathbb{R}$
- (iii) $\tan^{-1} e^x$ on $(0, \infty)$?
- (iv) $\tan^{-1} e^x$ on $(-\infty, 0)$
- (v) $\tan^{-1} e^x$ on $\mathbb{R}$

What are the points of inflection on the graph of $\tan^{-1} e^x$?

e^x and x^2 are both Convex

If $f: \mathbb{R}^+ \rightarrow \mathbb{R}$ is Concave

then $f(e^x)$, $f(x^2)$ may not be Convex. That is "Convexity can be lost by applying a Concave function to it"

Discuss this through several examples.

$\sec \theta$ is Convex and it retains its Convexity even after applying $\log (!)$

Polynomial Approximation

Taylor's Theorem: Before we state and prove this let us make a simple observation.

Assume $f: [a, b] \rightarrow \mathbb{R}$ cont and twice differentiable* on $[a, b]$. For $a < x < b$

$$f(x) = f(a) + (x-a) f'(c)$$

c between a and x

We can write (LMVT again) 

$$f'(c) = f'(a) + (c-a) f''(\xi)$$

for some $\xi \in (a, c)$

So

$$f(x) = f(a) + (x-a) f'(a) + (c-a)(x-a) f''(\xi)$$

Note that $|(c-a)(x-a)| \leq |x-a|^2$

So if x is close to a then,

$f(x) \doteq f(a) + (x-a) f'(a)$ is a good approximation since the "Error"

$$|(c-a)(x-a) f''(\xi)| \leq M |x-a|^2$$

$$M = \text{l.u.b. } |f''|$$

is small due to the ^{$[a, x]$} quadratic $|x-a|^2$ much

In fact we can do better than this

Instead of two c, ξ that arise ~~out~~ the

arise out of successive applications of
LMVT

* differentiable twice on an interval $(a-c, b)$. This is purely Motivational

-56-

We will obtain a formula with just
one single c - namely one single
application of Rolle's thm

Let's begin: Assume $f: I \rightarrow \mathbb{R}$ is twice
differentiable on an open interval
and $a, b \in I$ with $a < b$

$$\left(\begin{array}{c} \times \\ a \end{array} \quad \times \quad \begin{array}{c} \times \\ b \end{array} \right) \overset{I}{\rule{1.5cm}{0.4pt}}$$

Consider

$$\phi(x) = f(b) - f(x) - (b-x)f'(x)$$

$$\phi(b) = 0$$

But $\phi(a) \neq 0$. Let us modify $\phi(x)$

as

$$\underline{\Phi}(x) = \phi(x) - \left(\frac{b-x}{b-a} \right)^2 \phi(a)$$

$$\text{So } \underline{\Phi}(a) = 0 = \underline{\Phi}(b)$$

Rolle's thm can be applied. Let us
compute $\underline{\Phi}'(x)$:

$$\underline{\Phi}'(x) = \phi'(x) + \frac{2(b-x)\phi(a)}{(b-a)^2}$$

$$= -(b-x)f''(x) + \frac{2(b-x)\phi(a)}{(b-a)^2}$$

Now $\underline{\Phi}'(c) = 0$ for some $c \in (a, b)$

$$\therefore \phi(a) = \frac{1}{2} (b-a)^2 f''(c)$$

$$\therefore f(b) = f(a) + (b-a)f'(a) + \frac{1}{2} (b-a)^2 f''(c)$$

-57-

Now Suppose $f: I \rightarrow \mathbb{R}$ is thrice differentiable. Define

$$\phi(x) = f(b) - f(x) - (b-x)f'(x) - \frac{(b-x)^2}{2}f''(x)$$

$$\phi(b) = 0 \text{ but } \phi(a) \neq 0.$$

$$\text{So } \Phi(x) = \phi(x) - \left(\frac{b-x}{b-a}\right)^3 \phi(a)$$

$$\Phi(b) = \Phi(a) = 0.$$

Exercise: Prove that

$$f(b) = f(a) + (b-a)f'(a) + \frac{(b-a)^2}{2!}f''(a) + \frac{(b-a)^3}{3!}f'''(c)$$

Thm: (Taylor's Thm) If $f: I \rightarrow \mathbb{R}$ is n -times differentiable and $a, b \in I$
Then, $\exists$ c between a and b such that

$$f(b) = f(a) + (b-a)f'(a) + \dots + \frac{(b-a)^{n-1}}{(n-1)!}f^{(n-1)}(a) + R_n(a, b, c) \quad (*)$$

$$R_n(a, b, c) = \frac{(b-a)^n}{n!} f^{(n)}(c). \text{ Called the } n! \text{ Remainder in Taylor's}$$

Theorem - called Young's form of remainder.

Imp: c depends on n and b (for fixed a)

Remark: There are several different versions but this one suffices for our purpose

A great deal is known about the dependence of c on $a, b, \dots$. We shall not discuss.

Take $f(x) = \sin x$

$$f^{(k)}(x) = \sin\left(x + k\frac{\pi}{2}\right)$$

Take $a=0$: $f'(0)=1$, $f''(0)=0$, $f'''(0)=-1$

So (*) reads:

$$\begin{aligned} \sin b = & b - \frac{b^3}{3!} + \dots + (-1)^{n-1} \frac{b^{2n-1}}{(2n-1)!} \\ & + \frac{b^{2n}}{(2n)!} \sin(c + n\pi) \end{aligned}$$

$$R_n = \frac{b^{2n}}{(2n)!} \sin(c + n\pi)$$

Exercise: Prove that if $x \in \mathbb{R}$

$$\left| \frac{x^n}{n!} \right| \rightarrow 0 \text{ as } n \rightarrow \infty$$

So

$$\begin{aligned} \lim_{n \rightarrow \infty} \left(\sin b - \left(b - \frac{b^3}{3!} + \dots + (-1)^{n-1} \frac{b^{2n-1}}{(2n-1)!} \right) \right) \\ = \lim_n R_n = 0. \end{aligned}$$

$$\begin{aligned} \therefore \sin b = \lim_n \left(b - \frac{b^3}{3!} + \dots + (-1)^{n-1} \frac{b^{2n-1}}{(2n-1)!} \right) \\ = b - \frac{b^3}{3!} + \frac{b^5}{5!} - \dots \end{aligned}$$

Prove: $\cos b = 1 - \frac{b^2}{2!} + \frac{b^4}{4!} - \dots$

prove: $e^b = 1 + b + \frac{b^2}{2!} + \frac{b^3}{3!} + \dots$

Ex: -59-

$f(x) = \log(1+x) : \quad x \in (-1, \infty)$

Calculate the n th derivative of f at the origin.

Write down the Taylor series for f with remainder $R_n(b)$:

$$f(b) = f(0) + b f'(0) + \dots + \frac{b^{n-1} f^{(n-1)}(0)}{(n-1)!} + R_n(b)$$

Suppose $0 < b < 1$ Can you say $R_n(b) \rightarrow 0$?

Is it correct to say (Strict Ineq)

$$\log(1+b) = b - \frac{b^2}{2} + \frac{b^3}{3} - \dots ; \quad \underline{\underline{0 < b < 1}}$$

Is it correct to say

$$\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \dots$$

} Dangerous!

Result is correct-

but needs serious argument

Ex: Calculate the successive derivatives

of $(1+x)^{-1/2}$ at the origin and

Write out the Taylor series, $x \in (-1, \infty)$

Showing that $R_n(b) \rightarrow 0$ is going to involve some work!

Take $0 < b < 1$. Then $0 < c < b$

$0 < \frac{b}{c+1} < b$. Write out $R_n(b)$.

Would Stirling's formula help

in proving $R_n(b) \rightarrow 0$? Try it.

Que: What about things like $(1+x)^{-1/3}$ or $(1+x)^\alpha$? More troublesome.

Shall return to this later.

Maxima Minima again

Assume $f: (p-a, p+a) \rightarrow \mathbb{R}$ is k -times differentiable and the k -th derivative is Continuous. ($k \geq 4$)

(*) Assume $f'(p) = 0$, $f''(p) = 0$.

Further, (in addition to the above) at

(i) If $f'''(p) > 0$ then p is strictly increasing

If $f'''(p) < 0$ then, at p , f is strictly decreasing

In addition to $f'(p) = 0$, $f''(p) = 0$, (and $k \geq 5$)

(ii) If $f'''(p) = 0$ and $f^{(4)}(p) > 0$
 p is a point of local Min

If $f'''(p) = 0$ and $f^{(4)}(p) < 0$ the point p is a point of local Max

proof: Write down the Taylor series with

Remainder: Assume $f'''(p) > 0$.

$$f(x) - f(p) = \frac{f'''(p)(x-p)^3}{3!} + \frac{(x-p)^4}{4!} f^{(4)}(c)$$

c lies between p and x

$$\therefore \frac{f(x) - f(p)}{(x-p)^3} = \frac{f'''(p)}{3!} \left\{ 1 + \frac{(x-p)}{4} \frac{f^{(4)}(c)}{f'''(p)} \right\}, \quad (x \neq p)$$

Now if $f'''(p) > 0$; Look at the Expression

$$A(x) = \frac{(x-p)}{4} \frac{f^{(4)}(c)}{f'''(p)}. \quad \lim_{x \rightarrow p} A(x) = 0.$$

- 61 -

So in a small interval $(p-h, p+h) \subset (p-a, p+a)$
We have $-\frac{1}{2} < A(x) < \frac{1}{2}$.

$$\therefore \frac{1}{2} < 1 + A(x) < \frac{3}{2}$$

and $\frac{f(x)-f(p)}{(x-p)^3}$ lies between $\frac{1}{2} \frac{f'''(p)}{3!}$ and $\frac{3}{2} \frac{f'''(p)}{3!}$

So $f(x)-f(p)$ and $(x-p)$ have the same sign for $x \in (p-h, p+h)$

The other cases $f'''(p) < 0$ or
(ii) $f'''(p)$ is ALSO zero and $f^{(4)}(p) \neq 0$
are all left to the student.

Exercise: Let $f(x) = e^{-x^3} (1+x^4)^{10}$.

Is origin a point of Max/Min?
Discuss in detail.

Exercise: Let $y(x)$ be a function which is k times continuously diff.

with $k \geq 5$ such that

$$y'(x) = (y(x))^2 - 1 + \cos x$$

$$\text{and } y(0) = 0.$$

What can you say about the origin?

Exercise: Discuss what happens at origin to $f(x) = e^{-x^2} (1-x^2)^3$.